LLM-Finetuning

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LLM-Finetuning

What is Finetuning?

What

- Adjusting a pre-trained LLM to specialize in a specific domain or task
- uses task-specific datasets to update model weights

Why

Performance
enhances task-specific performance

Cost
cheaper than training from scratch

domain-specific
adapt to nuances and complexities

leverage existing knowledge of pre-trained models

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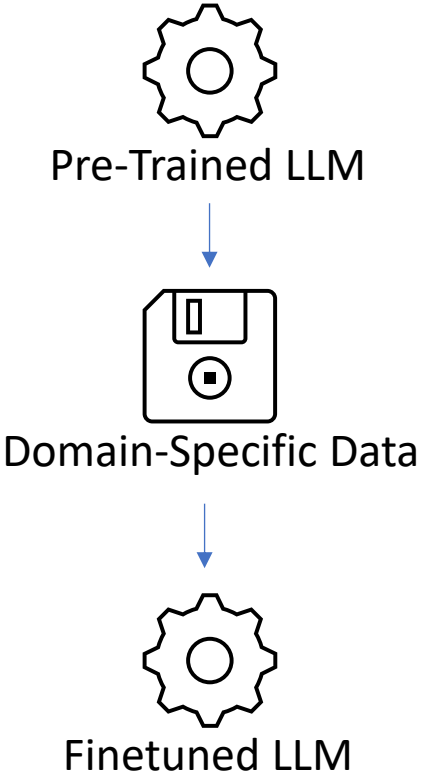
How?



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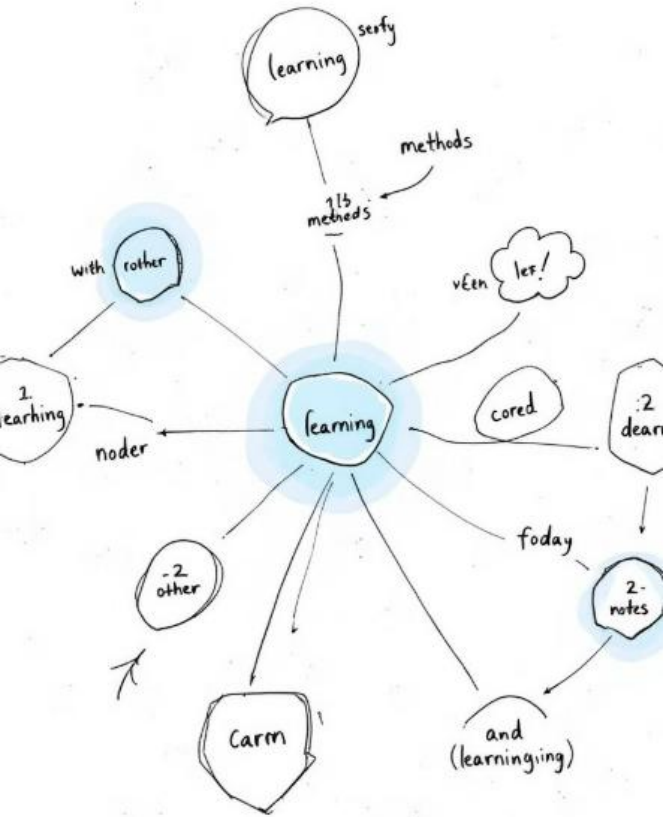
Methods



Supervised

using labeled data

Few-Shot

using limited examples

Domain

customize model to specific field

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Technical Implementation

Method	approach	pros	cons
Full Finetuning	train all parameters for pre-trained network	best performance, fully custom model	requires extensive computational resources, risk of catastrophic forgetting
Low-Rank Adaptation(LoRA) / Parameter efficient Finetuning (PEFT)	introduce low-rank matrices (small trainable layers), only train LoRA params	90% fewer params to train, faster, cheaper, avoid role of GPAI provider	less flexible

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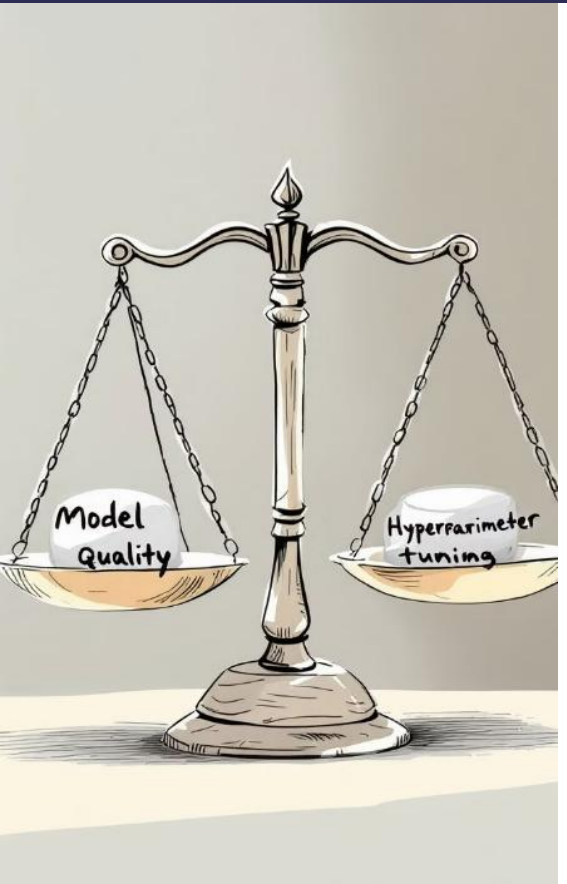
Challenges



- Ensure high Data Quality
- evaluate the model frequently
- prevent over- and underfitting
- balance specificity and generalization
- ethical considerations
- forgetting can occur
- high computational resources needed

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GPT-Finetuning

Fine-tuning

All | Successful | Failed

No fine-tuning jobs found

Create a fine-tuning job below or using the OpenAI API.

Learn more | + Create

Create a fine-tuned model

Method

Specify the method to be used for fine-tuning.

Supervised

- ✓ Supervised
- Direct Preference Optimization

Training data

Add a jsonl file to use for training. By providing the file, you confirm that you have the rights to use the data.

Upload new | Select existing

Upload a file or drag and drop here

(.jsonl)

Validation data

Add a jsonl file to use for validation metrics.

Upload new | Select existing | None

Suffix

Add a custom suffix that will be appended to the output model name.

my-experiment

Seed

The seed controls the reproducibility of the job. Passing in the same seed and job parameters should produce the same results, but may differ in

Learn about fine-tuning ↗ | Cancel | Create

https://platform.openai.com/finetune

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GPT-Finetuning

Direct Preference Optimization

- training uses comparisons between responses. given 2 responses (one preferred over the other), model learns to generate more preferred responses
- Example: chatbot
  - Q + 2A, human annotator prefers A over B → DPO adjusts model to increase likelihood of generating A over B
- goal: align better with user preference
- advantage over RLHF: more direct way to align models without needing complex RL

Supervised

- model trained on input-output pairs
- example: chatbot
  - dataset with Q and A
- goal: imitate response by minimizing difference between pred and ground truth
- limitation: does not directly optimize for user preference, just learns to replicate responses
- use supervised when high quality labeled data available

https://platform.openai.com/finetune

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GPT-Finetuning: Examples

Direct Preference Optimization

```
[
  {
    "input": "What is the capital of France?",
    "chosen": "Paris is the capital of France.",
    "rejected": "France's capital might be Paris, but I'm not sure."
  },
  {
    "input": "Explain the theory of relativity.",
    "chosen": "The theory of relativity, developed by Einstein,
describes the relationship between space and time...",
    "rejected": "Some scientists believe relativity is important, but
opinions vary."
  }
]
```

Supervised

```
[
  {
    "input": "What is the capital of France?",
    "output": "The capital of France is Paris."
  },
  {
    "input": "Explain the theory of relativity.",
    "output": "The theory of relativity, developed by Albert Einstein,
consists of two theories: special relativity and general relativity..."
  }
]
```

use when

- comparative feedback available
- you want to align model to user preference

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GPT-Finetuning: Comparison

Feature	Supervised	Direct Preference Optimization
Training Data	Labeled Input-output pairs	preference data
Training Objective	mimic given responses	prefer better responses
Risk of Bad Data	High (model copies bad responses)	lower (chooses better responses)
User Alignment	Weak	Strong
Complexity	simple	more sophisticated
	use when you want to finetune to domain-specific task	use when you want to align a model to user preferences